QUANTITATIVE LIPSCHITZ REGULARITY OF MOSER TRANSPORTS ALONG TEMPERING PATHS

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Abstract. We consider the construction of transport maps by Moser's method, with particular emphasis on obtaining explicit bounds on the Lipschitz constants of the transport. The basic construction is classical, prescribing a sufficiently regular path of probability measures, solving a family of time-varying Poisson equations to obtain a velocity field, and then integrating the corresponding ordinary differential equation.

The principal analytical question then concerns the differential regularity of the resulting velocity field. We show that for strongly log-concave measures with uniformly bounded Hessians, interpolated along a geometric tempering path, this control can be accomplished using only two elementary semigroup estimates.

1. The Moser–Poisson Construction

Let

$$\mu_t(dx) = \rho_t(x)\,dx, \qquad \rho_t(x) = Z_t^{-1} e^{-V_t(x)}, \qquad 0 \le t \le 1,$$

be a sufficiently regular path of probability measures on $\mathbb{R}^d$.

Define $\mathcal{L}_t = \Delta - \langle \nabla V_t, \nabla \rangle$ as the infinitesimal generator of the associated Langevin diffusions, and $g_t = \partial_t V_t - \mu_t(\partial_t V_t)$ as the local (in time) potential variation. Differentiating the density gives that $\partial_t \rho_t = -g_t \rho_t$. Define now u_t as the solution to the Poisson equation

$$\mathcal{L}_t u_t = g_t, \qquad \mu_t(u_t) = 0,$$

where the normalisation serves to fix the additive constant and otherwise has no effect on the subsequent construction.

Set $v_t = \nabla u_t$. Since $\nabla \cdot (\rho_t v_t) = \rho_t \mathcal{L}_t u_t = g_t \rho_t$, we deduce the continuity equation $\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0$; consequently, under the usual existence and regularity assumptions, the flow defined by

$$\frac{d}{dt} T_t(x) = v_t(T_t(x)), \qquad T_0(x) = x,$$

satisfies $(T_t)_{\#}\mu_0 = \mu_t$, and in particular, $T = T_1$ transports μ_0 onto μ_1.

The following elementary observation reduces quantitative regularity of the transport to derivative estimates for the solution of the Poisson equation.

Proposition 1. *(Regularity of the Moser flow) Suppose that the velocity fields $v_t = \nabla u_t$ are bounded and globally Lipschitz, with $A_t = \|\nabla u_t\|_\infty$, $B_t = \|D^2 u_t\|_\infty$, where A_t and B_t are integrable on $[0,1]$. Then the resulting Moser transport satisfies*

$$\|T - \mathrm{Id}\,\|_\infty \le \int_0^1 A_t\,dt, \qquad \mathrm{Lip}(T) \le \exp\left(\int_0^1 B_t\,dt\right).$$

Proof. The displacement estimate follows immediately from writing

$$|T_t(x) - x| \le \int_0^t |v_s(T_s(x))|\,ds \le \int_0^t A_s\,ds.$$

For $x, y \in \mathbb{R}^d$, the Lipschitz bound on v_t gives

$$\left|\frac{d}{dt}|T_t(x) - T_t(y)|\right| \le B_t |T_t(x) - T_t(y)|.$$

Grönwall's inequality therefore yields that $|T_t(x) - T_t(y)| \le e^{\int_0^t B_s\,ds}|x - y|$, and so the conclusion follows by taking $t = 1$. □

For the remainder of this note, we hence focus on establishing bounds on the gradient and Hessian of solutions to weighted Poisson equations, as this will allow control on the regularity of the map T.

2. Semigroup Estimates under Strong Convexity

Consider a potential $V : \mathbb{R}^d \to \mathbb{R}$ satisfying $D^2V \succeq mI_d$ for some $m > 0$. Write $\mathcal{L} = \Delta - \langle \nabla V, \nabla \rangle$ for the infinitesimal generator of the associated Langevin diffusion, and $P_s = e^{s\mathcal{L}}$ for the associated semigroup. The following lemma collects the two semigroup estimates that we shall need.

Lemma 2. *(Gradient contraction and resolvent smoothing) Suppose $D^2V \succeq mI_d$, with $m > 0$. Then*

(1) *For every Lipschitz function f,*
$$\|\nabla P_s f\|_\infty \le e^{-ms} \operatorname{Lip}(f).$$
(2) *For every bounded measurable function h and $s > 0$,*
$$\|\nabla P_s h\|_\infty \le \sqrt{\frac{m}{e^{2ms} - 1}}\, \|h\|_\infty.$$
(3) *For every bounded measurable function h, the resolvent $R_m = (m - \mathcal{L})^{-1} = \int_0^\infty e^{-ms} P_s \, ds$ satisfies*
$$\|\nabla R_m h\|_\infty \le \frac{1}{\sqrt{m}} \|h\|_\infty.$$

The resolvent estimate also holds for vector-valued functions, with Euclidean norm on the range and operator norm on the derivative.

Proof. The first estimate is the standard gradient contraction inequality under strong convexity. For example, it follows from synchronous coupling of the Langevin diffusion $dX_s = -\nabla V(X_s)\, ds + \sqrt{2}\, dB_s$, since trajectories driven by the same Brownian motion satisfy $|X_s^x - X_s^y| \le e^{-ms}|x - y|$ almost surely.

For the second estimate, recall the pointwise Bakry–Émery gradient inequality $|\nabla P_s f|^2 \le e^{-2ms} P_s |\nabla f|^2$ for strongly-convex V. Combined with the semigroup variance identity

$$P_s(f^2) - (P_s f)^2 = 2 \int_0^s P_r \left(|\nabla P_{s-r} f|^2 \right) \, dr,$$

one can deduce the 'reverse local Poincaré inequality'

$$P_s(f^2) - (P_s f)^2 \ge 2 |\nabla P_s f|^2 \int_0^s e^{2mr} \, dr = \frac{e^{2ms} - 1}{m} |\nabla P_s f|^2.$$

Since $P_s(f^2) - (P_s f)^2 \le \|f\|_\infty^2$, we hence obtain that

$$\|\nabla P_s f\|_\infty \le \sqrt{\frac{m}{e^{2ms} - 1}}\, \|f\|_\infty.$$

The estimate extends to bounded measurable functions by usual regularisation arguments.

Finally, integrating against the resolvent gives

$$\|\nabla R_m h\|_\infty \le \|h\|_\infty \int_0^\infty e^{-ms} \sqrt{\frac{m}{e^{2ms} - 1}}\, ds = \frac{\|h\|_\infty}{\sqrt{m}},$$

and so we conclude. For vector-valued h, apply the scalar estimate to the scalarisation $\langle h, e \rangle$ for arbitrary unit vectors e, and then take suprema. □

Lemma 3. *(Explicit Poisson derivative estimates) Let $V \in C^2(\mathbb{R}^d)$ satisfy*
$$mI_d \preceq D^2V(x) \preceq MI_d, \qquad x \in \mathbb{R}^d,$$
for constants $0 < m \le M < \infty$. Let $\mu(dx) = Z^{-1} e^{-V(x)}\, dx$, and suppose that g is Lipschitz, with $\mu(g) = 0$. Then, with $\mathcal{L} = \Delta - \langle \nabla V, \nabla \rangle$, the solution of $\mathcal{L}u = g$ satisfies
$$\|\nabla u\|_\infty \le \frac{\operatorname{Lip}(g)}{m}, \qquad \|D^2 u\|_\infty \le \frac{M}{m^{3/2}} \operatorname{Lip}(g).$$

Proof. Write $L = \operatorname{Lip}(g)$. Since g is centred, the Poisson solution admits the semigroup representation $u = -\int_0^\infty P_s g \, ds$. By gradient contraction, $\|\nabla P_s g\|_\infty \le L e^{-ms}$; integration with respect to s then gives the first claim.

To estimate the Hessian, set $w = \nabla u$, $H(x) = D^2V(x)$. Differentiating the Poisson equation once gives that $\mathcal{L}w - Hw = \nabla g$, where $\mathcal{L}$ is here taken to act componentwise on vector fields. Rearranging,

$$(m - \mathcal{L})w = (mI_d - H)w - \nabla g \implies w = R_m \left[(mI_d - H)w - \nabla g \right].$$

The preceding lemma therefore gives

$$\|D^2u\|_\infty = \|Dw\|_\infty \le \frac{1}{\sqrt{m}}\left(\|(mI_d - H)w\|_\infty + \|\nabla g\|_\infty\right).$$

By the two-sided Hessian bound, $\|mI_d - H(x)\| \le M - m$, and so the bound simplifies to

$$\|D^2u\|_\infty \le \frac{1}{\sqrt{m}}\left((M-m)\frac{L}{m} + L\right) = \frac{ML}{m^{3/2}}$$

as claimed. □

3. An Explicit Lipschitz Transport Estimate

We now apply the preceding estimates to the Moser construction.

Theorem 4. *Let $V_0, V_1 \in C^2(\mathbb{R}^d)$ satisfy*

$$mI_d \preceq D^2V_i(x) \preceq MI_d, \qquad x \in \mathbb{R}^d, \quad i = 0, 1,$$

for constants $0 < m \le M < \infty$. Suppose also that $L = \mathrm{Lip}(V_1 - V_0) < \infty$.

Define for $i \in \{0, 1\}$ the probability measures $\mu_i(dx) = Z_i^{-1}e^{-V_i(x)}\,dx$. Then Moser's construction, applied to the linear interpolation of potentials $V_t = (1-t)V_0 + tV_1$, produces a transport map T satisfying

$$T_\#\mu_0 = \mu_1, \qquad \|T - \mathrm{Id}\,\|_\infty \le \frac{L}{m}, \qquad \mathrm{Lip}(T) \le \exp\left(\frac{ML}{m^{3/2}}\right).$$

Proof. With V_t defined as above, form $\mu_t(dx) = Z_t^{-1}e^{-V_t(x)}\,dx$. Under the given assumptions, one checks quickly that $mI_d \preceq D^2V_t(x) \preceq MI_d$ uniformly in $t \in [0,1]$. Set $g_t = V_1 - V_0 - \mu_t(V_1 - V_0)$, and check that $\partial_t\mu_t = -g_t\mu_t$, $\mu_t(g_t) = 0$, and $\mathrm{Lip}(g_t) \le L$, again uniformly in $t \in [0,1]$. Writing $\mathcal{L}_t = \Delta - \langle\nabla V_t, \nabla\rangle$ for the infinitesimal generator of the associated Langevin diffusion, let u_t solve $\mathcal{L}_t u_t = g_t$, and define the Moser velocity $v_t = \nabla u_t$. The preceding Poisson estimate applies uniformly along the interpolation, whereby we obtain that $\|v_t\|_\infty \le L/m$, $\|Dv_t\|_\infty \le ML/m^{3/2}$.

Moreover, $\nabla\cdot(\mu_t v_t) = \mu_t\mathcal{L}_t u_t = g_t\mu_t$, so that $\partial_t\mu_t + \nabla\cdot(\mu_t v_t) = 0$, and hence the flow $\frac{d}{dt}T_t(x) = v_t(T_t(x))$, $T_0(x) = x$, satisfies $(T_t)_\#\mu_0 = \mu_t$, and $T = T_1$ indeed transports μ_0 onto μ_1.

Finally, our earlier regularity proposition gives that

$$\|T - \mathrm{Id}\,\|_\infty \le \int_0^1 \|v_t\|_\infty\,dt \le \frac{L}{m}, \qquad \mathrm{Lip}(T) \le \exp\left(\int_0^1 \|Dv_t\|_\infty\,dt\right) \le \exp\left(\frac{ML}{m^{3/2}}\right),$$

and we conclude. □

4. Remarks

4.1. **Interpolation Path.** The classical Dacorogna–Moser construction usually employs linear interpolation of densities, taking $\rho_t = (1-t)\rho_0 + t\rho_1$. Here we instead use linear interpolation of potentials, $V_t = (1-t)V_0 + tV_1$, or, equivalently, geometric interpolation of densities, $\rho_t \propto \rho_0^{1-t}\rho_1^t$. This choice is particularly convenient under strong convexity, because the uniform Hessian bounds are automatically preserved along the interpolation. The continuity-equation argument is the same in either case, but the corresponding Poisson equations differ.

4.2. **Hessian Bounds.** The lower bound $D^2V \succeq mI_d$ supplies the two semigroup estimates used in the proof. The upper bound $D^2V \preceq MI_d$ enters only through the differentiated Poisson equation, where it allows us to control $(mI_d - D^2V)\nabla u$. In this regard, the two sides of the assumption play different roles vis-à-vis contraction and smoothing properties.

This distinction becomes especially useful when extending the argument beyond the strongly convex setting. In particular, the first-order Poisson estimate can sometimes be obtained using reflection coupling, while resolvent smoothing may follow from local regularisation properties of the diffusion.

4.3. **Potential Regularity.** The semigroup and elliptic calculations may initially be justified for smooth potentials and forcing functions. The estimates themselves involve only the Lipschitz constant of the forcing function and the two-sided Hessian bounds on the potential.

Standard approximation and elliptic regularity arguments extend the result to the stated C^2 assumptions. In particular, smoothing the endpoint potentials with the same mollifier preserves the inequalities $mI_d \preceq D^2V_i \preceq MI_d$ and does not increase $\mathrm{Lip}(V_1 - V_0)$. The resulting estimates are therefore uniform under regularisation.

4.4. **Symmetry.** The main estimates described here concern the 'forward' transport T, but all assumptions are symmetric in μ_0, μ_1, and the Moser construction itself is symmetric under reversal of the interpolation. Consequently, the same estimates hold for the 'backward' transport T^{-1} also.